Q4 冷媒性能比较 / Refrigerant comparison

【1】逐图计算 / Calculations from the P-H diagrams

Refrigeration effect (q_e) = $h_1 - h_3$ (kJ/kg)

$$Coefficient\ \ of\ \ Performance\ \ (COP) = \frac{h_1 - h_3}{h_2 - h_1}$$

图 1 / Figure 1 — R-22

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 蓝色 / Blue	Room Cooler: 10℃ / 60℃ 橙色 / Orange
h_1 / kJ/kg	408	408.5
h_2 / kJ/kg	435	438
h_3 / kJ/kg	256	278
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	408 − 256 = 152	408.5 − 278 = 130.5
Compression work (w) h_2–h_1 / kJ/kg	435 − 408 = 27	438 − 408.5 = 29.5
Coefficient of Performance (COP)	$\frac{408 - 256}{435 - 408} = 5.63$	$\frac{408.5 - 278}{438 - 408.5} = 4.42$
Evaporating pressure p_e / bar(abs)	6	7
Condensing pressure p_c / bar(abs)	17	24.1
Specific volume (inlet) v_1 / m^3/kg	0.040	0.034
Specific volume (outlet) v_2 / m^3/kg	0.0155	0.0108
Entropy s / kJ/(kg·K)	1.75	1.73

图 2 / Figure 2 — R-407C

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 橙色 / Orange	Room Cooler: 10℃ / 60℃ 绿色 / Green
h_1 / kJ/kg	415	417
h_2 / kJ/kg	446	449
h_3 / kJ/kg	272	304
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	415 − 272 = 143	417 − 304 = 113
Compression work (w) h_2–h_1 / kJ/kg	446 − 415 = 31	449 − 417 = 32
Coefficient of Performance (COP)	$\frac{415 - 272}{446 - 415} = 4.61$	$\frac{417 - 304}{449 - 417} = 3.53$
Evaporating pressure p_e / MPa(abs)	0.55	0.64
Condensing pressure p_c / MPa(abs)	1.9	2.8
Specific volume (inlet) v_1 / m^3/kg	0.043	0.038
Specific volume (outlet) v_2 / m^3/kg	0.013	0.0082
Entropy s / kJ/(kg·K)	1.77	1.76

图 3 / Figure 3 — R-134a

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 绿色 / Green	Room Cooler: 10℃ / 60℃ 橙色 / Orange
h_1 / kJ/kg	401	404
h_2 / kJ/kg	426	434
h_3 / kJ/kg	263	287
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	401 − 263 = 138	404 − 287 = 117
Compression work (w) h_2–h_1 / kJ/kg	426 − 401 = 25	434 − 404 = 30
Coefficient of Performance (COP)	$\frac{401 - 263}{426 - 401} = 5.52$	$\frac{404 - 287}{434 - 404} = 3.90$
Evaporating pressure p_e / MPa(abs)	0.34	0.41
Condensing pressure p_c / MPa(abs)	1.2	1.7
Specific volume (inlet) v_1 / m^3/kg	0.060	0.053
Specific volume (outlet) v_2 / m^3/kg	0.017	0.014
Entropy s / kJ/(kg·K)	1.73	1.72

图 4 / Figure 4 — R-123

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 蓝色 / Blue	Room Cooler: 10℃ / 60℃ 橙色 / Orange
h_1 / kJ/kg	384	386
h_2 / kJ/kg	407	414
h_3 / kJ/kg	245	261
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	384 − 245 = 139	386 − 261 = 125
Compression work (w) h_2–h_1 / kJ/kg	407 − 384 = 23	414 − 386 = 28
Coefficient of Performance (COP)	$\frac{384 - 245}{407 - 384} = 6.04$	$\frac{386 - 261}{414 - 386} = 4.46$
Evaporating pressure p_e / MPa(abs)	0.04	0.051
Condensing pressure p_c / MPa(abs)	0.18	0.28
Specific volume (inlet) v_1 / m^3/kg	0.38	0.30
Specific volume (outlet) v_2 / m^3/kg	0.097	0.058
Entropy s / kJ/(kg·K)	1.67	1.66

【2】性能及构造比较 / Performance and construction comparison

（1）COP：两工况均 R-123＞R-22＞R-134a＞R-407C；同冷量下，COP 越高，压缩机功率及冷凝散热量越小。

Coefficient of Performance (COP): Both cases rank R-123 > R-22 > R-134a > R-407C. Higher COP reduces compressor power and condenser heat rejection at equal duty.

（2）制冷效应：R-22 两工况均最大，同冷量下质量流量最小。房间工况的 q_e、COP 均较低，压缩功 w 较大。

Refrigeration effect: R-22 has the greatest effect and lowest equal-duty mass flow. Room-cooler conditions give lower q_e/COP and greater w.

（3）蒸发压力：R-22 最高；R-123 最低且低于大气压，须良好密封、排不凝气体及控制水分。

Evaporating pressure: R-22 is highest; R-123 is lowest and sub-atmospheric, requiring good sealing, a purge unit and moisture control.

（4）冷凝压力：R-407C 最高、R-22 次高，设备须相应耐压及超压保护。R-407C 温度滑移须匹配换热器及 TEV 控制。

Condensing pressure: R-407C is highest, followed by R-22. Provide suitable pressure ratings and protection. R-407C glide affects heat exchangers and Thermal Expansion Valve (TEV) control.

（5）入口及出口比容：R-123 两处比容最大，同冷量下容积流量大，利于采用 Centrifugal compressor；R-22/R-407C 较利于紧凑型机组。

Specific volume: R-123 has the largest inlet/outlet specific volumes and equal-duty volume flow, favouring a Centrifugal compressor. R-22/R-407C favour compact units.

（6）熵：图示理想压缩取 $s_1=s_2$；实际绝热压缩产熵、耗功增大，COP 降低。不能凭绝对熵值给不同冷媒排优劣。

Entropy: The diagrams use $s_1 = s_2$. Real adiabatic compression generates entropy, increases work and lowers COP. Absolute entropy does not rank refrigerant efficiency.

【3】安全比较 / Safety comparison

（7）腐蚀性：四者均须相容的材料及润滑油，并保持干燥、洁净；R-123 尤其须防空气及水分吸入。

Corrosion: All four require compatible materials/lubricant and dry, clean systems. R-123 especially needs protection against inward air/moisture leakage.

（8）爆炸及爆裂：四者均为 1 级，标准试验下无火焰传播，但仍有超压爆裂风险，须压力切断及泄压保护。

Explosion: All four are class 1, with no flame propagation in classification tests. Overpressure can still rupture equipment; provide pressure cut-outs and relief protection.

（9）毒性：R-123 为 B1，毒性类别高于其余三者的 A1；四者均须控制泄漏及保持通风。

Toxicity: R-123 is B1, a higher toxicity class than A1 R-22/R-134a/R-407C. Control leaks and provide ventilation for all four.

【4】环境影响 / Environmental impacts

（10）ODP（臭氧层消耗潜能）：表示冷媒破坏臭氧层的能力，越低越好。R-22（0.05）及 R-123（0.014）会消耗臭氧；R-134a 和 R-407C 的 ODP 为 0，不会消耗臭氧层。

Ozone Depletion Potential (ODP): ODP measures damage to the ozone layer; lower is better. R-22 (0.05) and R-123 (0.014) deplete ozone; R-134a and R-407C have zero ODP.

（11）GWP（全球变暖潜能）：以同样质量的 CO_2 作比较，表示冷媒的增温影响，越低越好。按 100 年基准，R-22 为 1810、R-123 为 77、R-134a 为 1430、R-407C 约 1774。例如 1kg R-123 约相当于 77kg CO_2 的影响。因此 R-123 直接增温影响最低；四者都须减少泄漏，较高 COP 还能减少耗电引致的间接排放。

Global Warming Potential (GWP): GWP compares warming with the same mass of CO_2; lower is better. The 100-year values are R-22 1810, R-123 77, R-134a 1430 and R-407C about 1774. For example, 1kg of R-123 has about the impact of 77kg CO_2. R-123 has the lowest direct impact. Limit leakage; higher COP also reduces electricity-related emissions.

资料 / Sources: 手绘循环数据及 DuPont P–H 图；HVAC System II, slides 9–20, 29, 47–48；ASHRAE Handbook—Fundamentals (2025), Ch.29；IPCC AR4 (2007), Table 2.14（100-year GWP；R-407C 按质量加权 / mass-weighted）。